

Profiling & Alleviating Performance Bottlenecks in Zeek

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ABSTRACT

We present a proposal for identifying and alleviating bottlenecks in Zeek, a network security monitoring system. Prior work shows that CPU and memory consumption for Zeek, previously called Bro, depends on network traffic characteristics [2].

We outline a proposal for designing a reproducible profiling framework and using the resulting data to investigate strategies for mitigating bottlenecks. In this project, we will profile Zeek by testing different types of network traffic. Across these experiments, we will measure various performance metrics to identify a bottleneck. We will then design and evaluate a solution in its ability to improve performance.

This project relates to the course as it focuses on network traffic measurement, packet-processing bottlenecks, analysis of various flow types, and load balancing across workers.

1 PROBLEM & MOTIVATION

First developed under the name "Bro" in the 1990s, Zeek is an open-source network security monitoring platform that passively analyzes real-time network traffic, performing actions including protocol parsing, connection tracking, event generation, scripting, and logging. According to the Zeek Project website, it has over 10,000 deployments worldwide [5].

However, Zeek's packet-processing cost can vary substantially with traffic rate, packet size, flow churn, protocol mix, enabled analyzers, and scripts. Within a single Zeek worker, these costs can create processing bottlenecks, which may lead to incomplete visibility into network traffic as packets are dropped from processing queues. In deployments with multiple workers, which are typically assigned packets via flow hashing, this variable cost can also lead to load imbalances across workers. As a result, the actual processing capacity of a worker cluster may be lower than the theoretical maximum, as reflected in throughput and packet drops.

Zeek's users range from network security operators of enterprise networks to academic researchers. Our hope is to mitigate the bottlenecks that prevent them from using the tool and its underlying resources to the fullest potential. By improving on Zeek's packet-processing performance, its users will be able to analyze larger workloads with more accuracy and gain better visibility into network traffic.

2 RELATED WORK

The most foundational related work is *Bro: A System for Detecting Network Intruders in Real-Time* [4]. Though Zeek has since evolved, the paper highlights performance priorities, including maximizing throughput and avoiding packet drops. It identifies several techniques for improving performance, including filtering packets in the kernel and avoiding blocking the single thread. It also notes several limiting factors, including the handling of expired timers immediately, even during high load. Furthermore, to simplify the design, they chose to handle packet-related events immediately, a tradeoff that might have contributed to performance bottlenecks.

In *Re-examining the performance bottleneck in a NIDS with detailed profiling*, the authors profile Bro to guide future NIDS improvements [3]. While earlier work largely focused on speeding-up pattern matching, they note that improvements in pattern-matching did not necessarily translate into equivalent improvements in overall system performance, as other processing stages became bottlenecks. In their profiling of Bro, they identify policy script execution as a key limiting factor, leading to their recommendation of greater compilation into "binary codes native to the CPU."

Campbell and Lee take a different approach in *Prototyping a 100G Monitoring System* [1]. They go beyond profiling, attempting to reduce the amount of traffic that Bro analyzes. Their mitigation identifies large flows, implementing a "shunting" policy to limit packet processing once certain conditions are met. This reduces the load on Bro workers once enough information has been collected, demonstrating a mitigation targeted at one traffic type.

3 PROPOSED WORK AND EVALUATION

To begin, we will evaluate Zeek using controlled, reproducible workloads that vary four traffic characteristics: packet rate, packet size, flow churn, and protocol mix. Our assessment will focus on whether these workload characteristics contribute to processing bottlenecks or worker load imbalances. The data we collect may help explain why some flows are more expensive to process than others and which resources become limiting factors for performance.

Our main metrics will be:

- packet-processing throughput,

- packet-drop rate,
- CPU utilization,
- memory consumption, and
- worker load imbalance (for multi-worker deployments)

For each goal below, we will develop a mitigation for a bottleneck identified during profiling. We will then rerun the same workloads to identify the overhead introduced by the mitigation, using the same metrics.

3.1 75% Goal

Our first goal will be to profile Zeek and identify a single bottleneck associated with specific traffic characteristics. We will then implement a simple mitigation that addresses the identified bottleneck. For example, if a bottleneck was identified within a single worker, we could attempt to remove unnecessary processing. If a bottleneck regarding worker load imbalance was chosen, we could create a simple modification to Zeek's existing flow-assignment policy to distribute flows more evenly. We will then compare the proposed mitigation against Zeek's existing behavior under the same workload.

Evaluation: This goal first entails the creation of a Zeek benchmarking environment, with reproducible results for various traffic workloads. We should identify a performance bottleneck, then develop and implement a simple mitigation. This mitigation will be considered successful if it demonstrates measurable performance improvements in any of the aforementioned evaluation metrics.

3.2 100% Goal

Next, we will design and implement an improved mitigation for the bottleneck identified during profiling, based on 75% evaluation results. Going beyond optimizing for a single motivating workload, our mitigation will strive to be reasonably performant across most traffic workloads.

For example, for worker load imbalance, we might implement a cost-aware flow-assignment strategy that considers a new flow's estimated processing cost and the current load of each worker, while still ensuring flow affinity.

Evaluation: This stage will be assessed across multiple traffic mixes and rates against unmodified Zeek and the 75% implementation, using the metrics above. We will also characterize the mitigation's overhead and limitations.

3.3 125% Goal

Our 125% goal extends the 100% solution to dynamically analyze and respond to changing workloads.

For worker imbalance, one possible extension is to predict a new flow's processing cost using information from its first few packets (e.g., packet size, header metadata, transport protocol). Another option is to dynamically scale the number of Zeek workers or possibly migrate flows from one worker to

another. For other bottlenecks, possible ideas include adapting to changing protocol mixes and/or targeting unusually expensive flows.

Evaluation: This stage will be evaluated by its benefit (and cost) in comparison to the 100% solution and unmodified Zeek implementation. The 125% solution should demonstrate the ability to determine changing workload conditions and adjust accordingly. This solution will be evaluated by the extra performance it provides relative to the cost of the dynamic reasoning.

4 REQUIRED RESOURCES

No specialized hardware is needed. Our main resources are:

- **CloudLab** for compute resources
- **tcpreplay** for existing packet captures
- **iperf3** for varying packet sizes and flow patterns
- **DPDK** for high-speed traffic generation
- **Linux perf** for profiling CPU usage
- **zeekctl top** for per-worker CPU/memory usage
- **Zeek's Prometheus tool** for metric logging

5 TEAM MEMBER RESPONSIBILITIES

We will meet in person weekly to set goals and develop designs together, then complete our assigned work before regrouping. Finally, we will check in periodically with Dr. Sengupta for feedback and to identify missing elements.

We will each pick one or two of the packet characteristics to profile, collecting the data required for the remaining parts of the project. Together, we will then identify the bottleneck that appears most promising. For each milestone, we will investigate potential mitigation strategies, combine our ideas into a single solution, and implement it together.

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